

Increasing Play-Related Statements in Children with Autism Toward Their Siblings: Effects of Video Modeling

Bridget A. Taylor,^{1,2} Len Levin,¹ and Suzanne Jasper¹

Two experiments were conducted to assess the effects of video modeling procedures to increase play comments in two children with autism toward their siblings. A multiple baseline probe design was used in both studies to assess the intervention across three play activities for each participant. Study 1 used video modeling procedures similar to Charlop and Milstein (1989). The participant viewed scripted play comments between his sibling and an adult, and then participated in practice sessions with the same stimuli and an adult. Probe sessions assessed play comments with the sibling, in the absence of video viewing. Results of this study revealed the participant learned to make the scripted play comments for all three play activities. In Study 2, a forward chaining procedure was implemented to assess acquisition of a longer series of comments, as well as the effects of video models that did not contain scripted comments by the participant's sibling. The participant viewed brief segments of video models of play comments between an adult and the participant's sibling in a sequential format and participated in practice sessions with an adult with the same stimuli. Retention probes were conducted to assess play comments with the sibling in the absence of video viewing. Results of this study indicated that the participant made a higher number of play comments following video modeling intervention than prior to intervention for all three play activities. Additionally, comments were both scripted and unscripted for this participant. Both of these studies revealed that video modeling was an effective intervention for teaching children with autism to make play comments toward their siblings.

KEY WORDS: autism; play-related statements; multiple baseline probe.

¹Alpine Learning Group.

²To whom correspondence should be addressed at 777 Paramus Road, Paramus, New Jersey 07652.

INTRODUCTION

A challenging task for educators of children with autism is to develop interventions that lead to engagement in reciprocal social interaction. Children with autism are deficient in their ability to engage in joint attention concerning play activities, initiating conversation, and responding to initiations made by their peers (American Psychiatric Association, 1994). However, a number of studies have documented that children with autism can be taught to engage in socially relevant responses such as initiating conversation (Krantz and McClannahan, 1993), asking questions (Taylor and Harris, 1995), and making statements of affection (Charlop and Walsh, 1986).

Researchers and clinicians have demonstrated the effects of a number of interventions to teach social language skills to individuals with autism. These include use of textual prompts (Krantz and McClannahan, 1993; Matson *et al.*, 1993) time delay procedures (Charlop and Walsh, 1986), priming (Zanolli *et al.*, 1996) incidental teaching (McGee *et al.*, 1992), pivotal response training (Peirce and Schreibman, 1995), and tactile prompts (Taylor and Levin, 1998). All of these studies revealed that individuals with autism can learn to initiate some form of social interaction with peers.

One procedure, video modeling, has been demonstrated to be effective to teach children with autism simple exchanges of conversation (Charlop and Milstein, 1989). In this study, three children were taught to engage in conversational exchanges about objects (e.g., toys) with an adult. Children watched video taped scripted conversations between two adults and were then presented with the same stimuli. Charlop and Milstein documented that following a number of video exposures, all three children learned the scripted conversation. In addition, unscripted statements increased for all of the participants.

The purpose of the present studies was to assess the effects of video modeling to teach two children with autism to engage in play comments with their sibling in the home setting. The current studies extend Charlop and Milstein's study in several ways. First, intervention and probe sessions were conducted in the students' homes as opposed to a treatment center. Second, siblings served as one of the models in the video, as opposed to two adults. Third, children were taught to respond and initiate to their sibling, as opposed to an adult. In addition, Study 2, used a variation of Charlop and Milstein's protocol by using a forward chaining procedure to assess acquisition of a longer sequence of verbal comments.

GENERAL METHOD

Participants

Both participants were students enrolled in the Alpine Learning Group, a center-based education program for children with autism, and met the criteria for autistic disorder according to the *Diagnostic and Statistic Manual* (American Psychiatric Association, 1994). Jack participated in Experiment 1; he was six years old at the time of the study. His eight year old sister Shari participated in the video modeling and play sessions. Jack was able to label nouns and verbs, imitate responses demonstrated by adults, request in three- to four-word sentences, and complete a number of academic tasks (e.g., site word recognition, writing, and addition). He played appropriately with a number of toys but was not observed to play with his sibling or make play comments toward her.

Manuel, the participant in Experiment 2, was nine years old at the time of the study. His six year old brother Brian participated in the video modeling and play sessions. Manuel spoke in seven to ten word sentences, participated in academic activities (e.g., reading, writing, and math), and was observed to engage in simple exchanges of conversation. His comments, however, were often unrelated to the activity and not socially relevant (e.g., asking about street addresses, talking about dates). Although Manuel was observed to play along side his brother during certain play activities, he rarely initiated play or made appropriate play comments.

Setting

Video viewing, practice sessions and retention probes all took place in the homes of the participants. Video viewing took place in the family room of each house. Baseline, practice and retention probes took place in either the play room or the participants' bedrooms.

Dependent Measures

The dependent measures were the percentage of scripted comments for Jack, and the number of scripted and unscripted play comments for Manuel. Scripted comments were defined as those comments viewed on the video tape. Unscripted comments were any play related comments presented in a full sentence and were appropriate to the context (e.g., "This car goes fast!"). Repetitions of prior comments were not scored.

Preteaching with Sibling

Jack and Manuel's respective siblings were asked by their parents if they wanted to help teach their sibling how to talk about their play activities. Both agreed to participate and appeared to be interested in playing with their sibling. Preteaching sessions were conducted to review the video taping procedure with the siblings, and in the case of Jack's sister Shari, to review the script she was to read for the video tape. Role play exercises were used along with verbal praise to teach siblings how to engage in the play activities for the video taping and the play sessions. In the case of Manuel's brother, he was told to talk and play as he typically did, but if Manuel talked to him, he should talk back. Prior to each experimental session, siblings were asked if they wanted to play with their respective brothers. If a sibling chose not to participate, then a probe session (i.e., play session) was not conducted on that day.

Stimulus Materials

Play materials were chosen based on participant and sibling interest (e.g., a car track set and color forms). Normative samples of play comments were derived prior to making the video segments by having an age-matched peer play with each toy along with an adult. During these sessions, the adult recorded all comments made by the peer about the play activity. A sample of the comments was then chosen for the video segments.

Length of comments was determined individually for each participant. In the case of Jack, direct assessment indicated that he could only imitate four to five word sentences. Therefore, length of comments for Jack was three to four words (e.g., "The airplane flies."). Manuel on the other hand was able to imitate six to seven word sentences so the comments on his video segments were longer ("You be the bad guys and I'll be the good guy!").

During taping of the video for Experiment 1, play materials for each video were presented and the sibling (Shari) was provided with a script to read containing play comments or requests (e.g., "Let's play with the trains."). An adult sat next to the sibling and read an alternate script containing the comments derived from the normative sample, and manipulated the toys according to the script (e.g., handed the train to the sibling). Three different tapes (one for each play activity) were made, each containing an average of six play comments for the participant to imitate.

In the taping of the video for Experiment 2, an adult read scripted comments relevant to the play activities (e.g., "This is the bad guy!"), but the sibling was not provided with a script and whatever comments the

sibling happened to make were recorded. Three tapes were made each containing an average of 10 comments for the participant to imitate. The purpose of not providing a script to the sibling for this experiment was to assess the effects of video modeling when the alternate responses were not presented in a rote, scripted fashion. As in Experiment 1, it was anticipated that the participant would imitate the comments made by the adult, since play sessions would be conducted in the presence of the sibling.

Experimental Design

Both studies used a multiple baseline probe design across 3 play activities. In Experiment 1, intervention probes were conducted prior to each video modeling intervention session to assess play comments in the absence of video viewing. In Experiment 2, intervention probes were conducted before the daily intervention session, but only after Manuel had completed the forward chain procedure (i.e., met criteria with an adult on each step by saying the same number of statements that were on the video segment during practice sessions that occurred immediately after viewing the video segment).

Interobserver Agreement

Interobserver agreement was scored during 30% and 40% of the baseline sessions and during 50% and 40% of the probe sessions for Jack and Manuel, respectively. Interobserver agreement was calculated by dividing the total number of agreements by total agreements plus disagreements and multiplying by 100%. Mean agreement for baseline, for both participants was 100%. Mean agreement for probe sessions, for both participants was 97% (range of 70% to 100%).

EXPERIMENT 1

Procedure

Baseline

During baseline, Jack and his sister were provided with each play activity and the instruction "Play together." Jack's sister stated her portion of the play script and paused after each statement to provide an opportunity

for Jack to comment back (e.g., “These hotdogs taste yummy.”). Data collectors recorded Jack’s responses and any independently generated comments. Each play session lasted 5 min and one to three play sessions were conducted daily.

Video Modeling Intervention

During intervention sessions, Jack viewed the entire video segment three consecutive times. He was then provided with the same stimuli related to the video segment and an adult read Shari’s portion of the script. If Jack stated any of the scripted play comments from the video tape, the adult provided verbal praise and a tangible reward (e.g., an edible). If Jack did not state any of the portions of his script, the adult continued to present Shari’s scripted statements. Jack then viewed the segment one additional time and practiced with the adult.

Probe Sessions

Probes were run each day, prior to any video modeling intervention and practice sessions. As in the baseline sessions, Jack and Shari were seated at the table or on the floor and were provided with the toys and the instruction “Play together.” Shari read her portions of the script and paused following each comment to provide Jack with an opportunity to comment back. If he did not, she continued to state her portion of the script. Verbal praise and tangible rewards were not provided during probe sessions. Video modeling intervention and practice sessions were conducted immediately after these probe sessions.

Results and Discussion

Figure 1 depicts percentage of play comments stated per probe session across all three play activities. In baseline, Jack did not state any play comments (scripted or unscripted). Following the video modeling intervention Jack learned to state most of the play comments depicted on the video. Within seven probe sessions, Jack met the 80% criterion on the “Airplane” comments and maintained performance with a mean of 94%. Jack’s performance was somewhat more variable for the “Picnic” comments. He met criterion within nine sessions and maintained performance at 70%. The “Dinosaur Cars” comments were mastered the fastest; within four sessions.

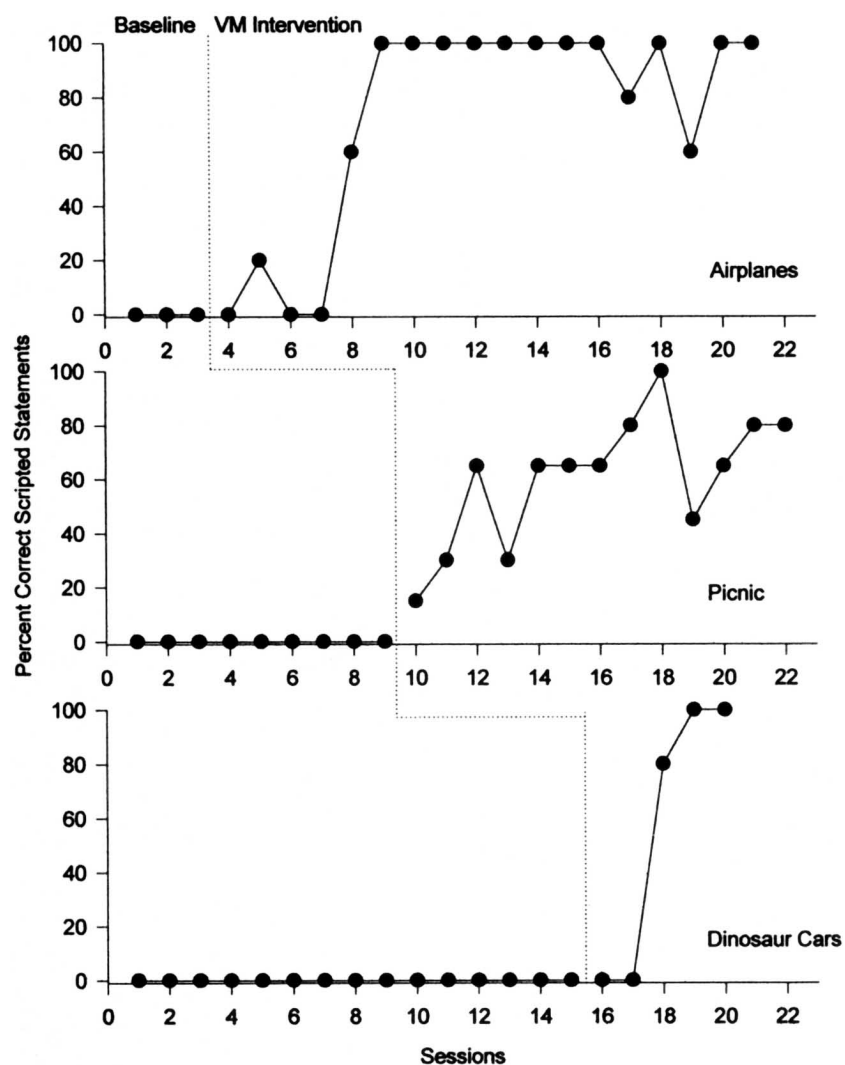


Fig. 1. The percentage of scripted comments stated during baseline and probe sessions for Experiment 1 during all three play activities.

There were no unscripted comments recorded in baseline or during video modeling intervention.

Results of the multiple baseline revealed that the video modeling intervention was effective in teaching Jack to make the scripted play comments to his sister during the three play activities. The intervention, how-

ever, did not lead to demonstration of unscripted comments. It may be that for some children with autism, video modeling intervention will be effective to teach scripted conversation only. For these children additional instructional strategies may be warranted to teach unscripted comments.

EXPERIMENT 2

Procedure

Baseline

During baseline, Manuel and his brother were provided with each play activity and the instruction "Play together." Data collectors recorded play related comments made by Manuel. Each play session lasted 5 min and one to three play sessions were conducted daily.

Video Modeling Intervention

Initially, during intervention sessions, Manuel viewed the first four comments on the video and was then provided with the same stimuli related to the video segment. An adult sat near Manuel, played with the toys, and stated unscripted comments relevant to the play activity. If Manuel stated any scripted, or unscripted play comments from the video tape, the adult provided verbal praise (e.g., "I like the when you talk about your toys!"), and a tangible reward (e.g., an edible). Over the course of the intervention sessions, the length of the segment viewed on the tape increased as Manuel's performance in stating play comments (i.e., scripted and unscripted) increased during the practice sessions.

Probe Sessions

Probes were conducted after Manuel completed the forward chain procedure and met acquisition criteria during practice sessions. To satisfy acquisition criteria, Manuel had to produce at least the same number of play statements (i.e., scripted or unscripted) as were viewed on the video segment. For example, if, as part of the forward chain procedure, Manuel viewed the first four statements, performance criterion for the practice session immediately following the viewing was four statements. That is, if Manuel independently generated at least four play statements, he moved

to the next step of the forward chain procedure (e.g., viewing the first six play statements on the video). Mean number of sessions needed until Manuel met acquisition criteria on all three play activities was 7 (i.e., range was 6–8).

During probe sessions, as in the baseline sessions, Manuel and his brother were provided with the toys and the instruction “Play together.” Data collectors recorded any play comments exhibited by Manuel. Verbal praise and tangible rewards were not provided during probe sessions.

Results and Discussion

Figure 2 depicts number of scripted and unscripted comments stated per probe session across all three play activities. In baseline, the frequency of play comments were relatively low as compared to probe sessions. Mean number of statements per session during baseline was 0, 2.5 and .6 for each play activity respectively. Mean number of statements per session during intervention probe sessions was 8.3, 9.4 and 8.6, respectively for each play activity.

Figure 3 displays mean number of scripted and unscripted comments for each play activity. Mean occurrence of unscripted comments was 3, 6, and 3 for batman, cars and marines, respectively. Mean number of unscripted comments surpassed the number of scripted for only one activity, cars.

Results of this study indicate that a forward chaining video modeling procedure can lead to an increase in play comments. Manuel learned to comment toward his sibling about his play activities via the video modeling intervention. Additionally, in this experiment, the sibling did not provide scripted comments during the probes sessions (i.e., the sibling did not use the same comments that were used on the video segment). This suggests that Manuel’s comments were evoked by the play materials or by presence of his sibling; stimuli that occur more naturally in the environment than specific scripted statements.

GENERAL DISCUSSION

Children with autism require systematic intervention to learn to engage in reciprocal social interaction. Video modeling (Charlop and Milstein, 1989) is one intervention that has been effective in teaching children with autism to engage in simple conversational responses. These two studies indicated that similar procedures can be used to teach children with autism

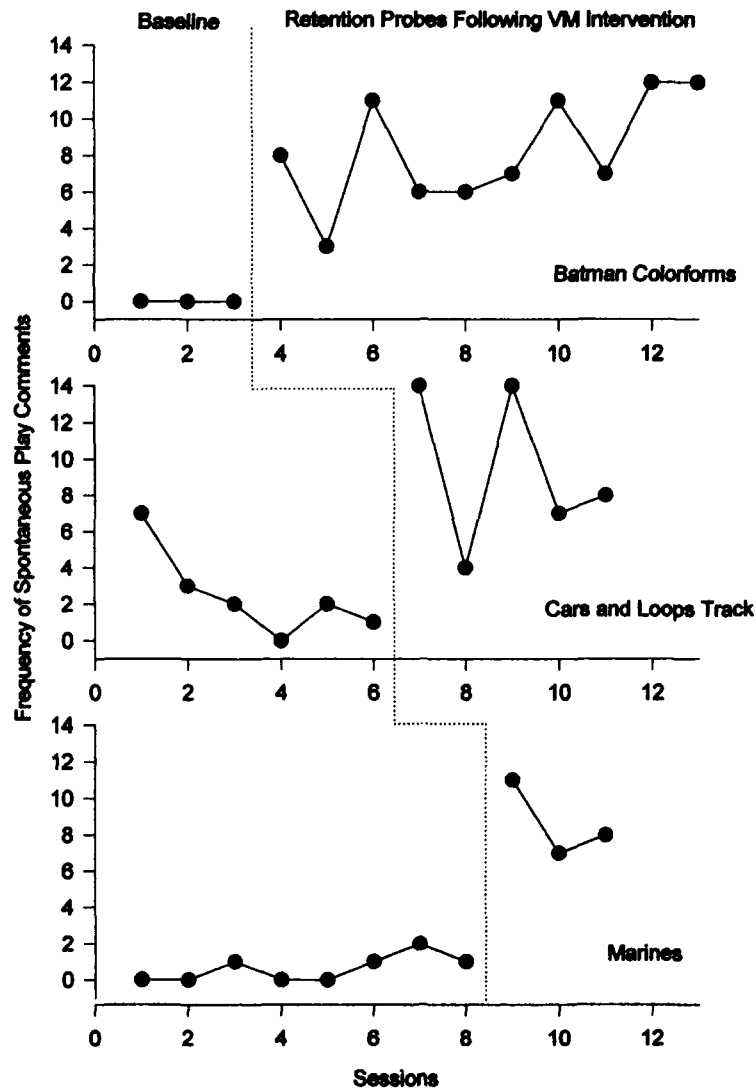


Fig. 2. The number of scripted and unscripted comment stated during baseline and probes sessions for Experiment 2 during all three play activities.

to engage in play comments with siblings in the home setting. In addition, procedures in Experiment 2, indicated that children can learn to make considerably longer play comments with a forward chain procedure, and can make these comments in the presence of unscripted verbal stimuli.

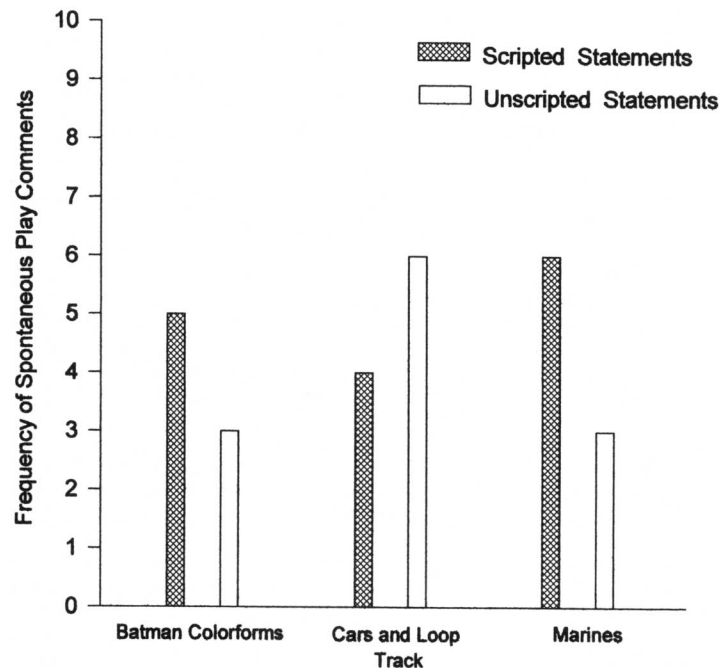


Fig. 3. The mean number of scripted and unscripted comments stated during the three activities.

Further, Experiment 2, led not only to an increase in the scripted comments for Manuel, but also to unscripted comments.

One limitation of these studies is that the different outcomes of the two experiments (i.e., Manuel learned to make scripted and unscripted statements while Jack only displayed an increase in scripted statements) can not necessarily be attributed to the different procedures used. One could speculate that the variety of play comments made by Manuel's brother (i.e., since the comments were not scripted, Brian's comments varied from session to session), facilitated the production of a greater variety of comments (i.e., unscripted) by Manuel. However, the expressive language abilities of the two participants were significantly different. Consequently, the different results may reflect individual differences rather than differential effects of the two interventions. Future studies may want to use participants with similar expressive language abilities to assess the differential outcomes of these video modeling interventions. Additionally, language assessment procedures may be necessary to determine which children would benefit most from various video modeling procedures.

REFERENCES

- American Psychiatric Association (1994). *Diagnostic and Statistical Manual of Mental Disorders* (4th ed.), Washington, DC: Author.
- Charlop, M. H., and Walsh, M. E. (1986). Increasing autistic children's spontaneous verbalizations of affection: An assessment of time delay and peer modeling procedures. *J. Appl. Behav. Anal.* 19: 307-314.
- Charlop, M. H., and Milstein, J. P., (1989). Teaching autistic children conversational speech using video modeling. *J. Appl. Behav. Anal.* 23: 275-285.
- Krantz, P. J., and McClannahan, L. E. (1993). Teaching children with autism to initiate to peers: Effects of a script fading procedure. *J. Appl. Behav. Anal.* 26: 121-132.
- Krantz, P. J., MacDuff, M. T., and McClannahan, L. E. (1993). Programming participation in family activities for children with autism: Parents use of photographic schedules. *J. Appl. Behav. Anal.* 26: 137-138.
- Matson, J. L., Sevin, J. A., Box, M. L., Francis, K. L., and Sevin, B. M. (1993). An evaluation of two methods for increasing self-initiated verbalizations in autistic children. *Journal of Applied Behav. Anal.* 26: 389-398.
- McGee, G. G., Almeida, M. C., Sulzer-Azaroff, B., and Feldman, R. S. (1992). Promoting reciprocal interactions via peer incidental teaching. *J. Appl. Behav. Anal.* 25: 515-524.
- Peirce, K., and Schreibman, L. (1995). Increasing complex social behavior in children with autism: Effects of a peer mediated pivotal response training. *J. Appl. Behav. Anal.* 28: 285-295.
- Taylor, B. A., and Harris, S. L. (1995). Teaching children with autism to seek information: Acquisition of novel information and generalization of responding. *J. Appl. Behav. Anal.* 28: 3-14.
- Taylor, B. A., and Levin, L. (1998). Teaching a student with autism to make verbal initiations: Effects of a 'tactile prompt' device. *J. Appl. Behav. Anal.* 31: 651-654.
- Zanolli, K., Daggett, J., and Adams, T. (1996). Teaching preschool age autistic children to make spontaneous initiations to peers using priming. *J. Aut. Devel. Dis.* 26: 407-422.